KL Divergence Between Gaussians

the case we use in the lectures

Introduction to Generative AI in Computer Vision

Supplementary note

Why this note. Two places in the course need the KL divergence between two Gaussians, and in both the lecture states the answer without proving it:

- **Lecture 1 (VAE).** The KL term of the training loss becomes the per-coordinate sum $\frac{1}{2}\sum_j\big[\mu_j^2 + \sigma_j^2 - \log\sigma_j^2 - 1\big]$.
- **Lecture 2 (DDPM).** The KL between the true and the learned reverse step collapses to a *squared error*, which is the reason training a diffusion model is a regression problem.

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1 Two independent problems become one

Lemma 1 (KL adds over independent coordinates)**.** *Suppose q and p both factorise over the coordinates of $\mathbf{x} = (x_1, \dots, x_d)$, that is $q(\mathbf{x}) = \prod_{j=1}^d q_j(x_j)$ and $p(\mathbf{x}) = \prod_{j=1}^d p_j(x_j)$. Then*

$$\mathcal{D}_{\mathrm{KL}}(q \,\|\, p) = \sum_{j=1}^{d} \mathcal{D}_{\mathrm{KL}}(q_j \,\|\, p_j). \tag{1}$$

Proof. A product inside a logarithm becomes a sum:

$$\log\frac{q(\mathbf{x})}{p(\mathbf{x})} = \log\prod_{j=1}^{d}\frac{q_j(x_j)}{p_j(x_j)} = \sum_{j=1}^{d}\log\frac{q_j(x_j)}{p_j(x_j)}.$$

Now take the expectation over $\mathbf{x} \sim q$ and use linearity. The j-th summand depends on x_j only, and the marginal of x_j under q is q_j, so its expectation is $\mathbb{E}_{x_j\sim q_j}[\log(q_j/p_j)] = \mathcal{D}_{\mathrm{KL}}(q_j\|p_j)$. □

A Gaussian with diagonal covariance $\boldsymbol{\Sigma} = \operatorname{diag}(\sigma_1^2, \dots, \sigma_d^2)$ factorises exactly this way, since its density is a product of d one-dimensional Gaussian densities. So by Lemma 1 it is enough to solve the problem for $d = 1$.

2 The scalar case

Proposition 1 (Scalar Gaussian KL)**.** *For $q = \mathcal{N}(\mu_q, \sigma_q^2)$ and $p = \mathcal{N}(\mu_p, \sigma_p^2)$ with $\sigma_q, \sigma_p > 0$,*

$$\mathcal{D}_{\mathrm{KL}}(q \,\|\, p) = \log\frac{\sigma_p}{\sigma_q} + \frac{\sigma_q^2 + (\mu_q - \mu_p)^2}{2\sigma_p^2} - \frac{1}{2}. \tag{2}$$

Proof. Write out the two log-densities,

$$\log q(x) = -\tfrac{1}{2}\log\big(2\pi\sigma_q^2\big) - \frac{(x-\mu_q)^2}{2\sigma_q^2}, \qquad \log p(x) = -\tfrac{1}{2}\log\big(2\pi\sigma_p^2\big) - \frac{(x-\mu_p)^2}{2\sigma_p^2},$$

and subtract. The 2π's cancel, leaving

$$\log \frac{q(x)}{p(x)} = \log \frac{\sigma_p}{\sigma_q} - \frac{(x-\mu_q)^2}{2\sigma_q^2} + \frac{(x-\mu_p)^2}{2\sigma_p^2}. \tag{3}$$

Since $\mathcal{D}_{\mathrm{KL}}(q\|p) = \mathbb{E}_{x\sim q}\big[\log \frac{q(x)}{p(x)}\big]$, we only need the average of the two squares under q.

First square. By definition of the variance, $\mathbb{E}_q\big[(x-\mu_q)^2\big] = \sigma_q^2$, so the middle term of (3) averages to $-\sigma_q^2/(2\sigma_q^2) = -\frac{1}{2}$.

Second square. Here x is spread around μ_q, not μ_p. Add and subtract μ_q so that the random part is centred:

$$(x-\mu_p)^2 = \big[(x-\mu_q) + (\mu_q-\mu_p)\big]^2 = (x-\mu_q)^2 + 2(\mu_q-\mu_p)(x-\mu_q) + (\mu_q-\mu_p)^2.$$

Average term by term: the first gives σ_q^2; the middle one vanishes because $\mathbb{E}_q[x-\mu_q]=0$; the last is a constant. Hence

$$\mathbb{E}_q\big[(x-\mu_p)^2\big] = \sigma_q^2 + (\mu_q-\mu_p)^2, \tag{4}$$

the familiar "variance plus squared bias".

Substituting both averages into (3) gives (2). □

3 The two cases the lectures use

Corollary 1 (VAE training loss). *Let $q = \mathcal{N}\big(\boldsymbol{\mu}, \mathrm{diag}(\sigma_1^2,\ldots,\sigma_d^2)\big)$ and let $p = \mathcal{N}(\mathbf{0},\mathbf{I})$ be the standard prior. Then*

$$\mathcal{D}_{\mathrm{KL}}(q\,\|\,p) = \frac{1}{2}\sum_{j=1}^{d}\Big[\mu_j^2 + \sigma_j^2 - \log\sigma_j^2 - 1\Big]. \tag{5}$$

Proof. Both distributions are diagonal, so Lemma 1 applies and we may work one coordinate at a time. In coordinate j we have $\mu_q=\mu_j$, $\sigma_q=\sigma_j$, and from the prior $\mu_p=0$, $\sigma_p=1$. Then (2) reads

$$\mathcal{D}_{\mathrm{KL}}(q_j\|p_j) = \underbrace{\log\frac{1}{\sigma_j}}_{=-\frac{1}{2}\log\sigma_j^2} + \frac{\sigma_j^2+\mu_j^2}{2} - \frac{1}{2} = \frac{1}{2}\Big[\mu_j^2 + \sigma_j^2 - \log\sigma_j^2 - 1\Big].$$

Summing over j gives (5). □

Remark. Because the coordinates never interact, the VAE's KL term is a plain sum with no cross terms — which is why it can be implemented in one line. Reading a single summand also shows the two separate pressures it applies: μ_j^2 is smallest at $\mu_j=0$, and $\sigma_j^2 - \log\sigma_j^2 - 1$ is smallest at $\sigma_j^2=1$, where it equals exactly 0. So the term pushes the posterior towards the prior and vanishes when they agree.

Corollary 2 (Diffusion training loss). *Let $q = \mathcal{N}(\boldsymbol{\mu}_q, \sigma^2\mathbf{I})$ and $p = \mathcal{N}(\boldsymbol{\mu}_p, \sigma^2\mathbf{I})$ share the* same *isotropic covariance. Then*

$$\mathcal{D}_{\mathrm{KL}}(q\,\|\,p) = \frac{\|\boldsymbol{\mu}_q - \boldsymbol{\mu}_p\|_2^2}{2\sigma^2}. \tag{6}$$

Proof. Again both are diagonal, so work coordinate by coordinate with $\sigma_q = \sigma_p = \sigma$ in (2). The logarithm is $\log 1 = 0$, and

$$\mathcal{D}_{\mathrm{KL}}(q_j \| p_j) = \frac{\sigma^2 + (\mu_{q,j} - \mu_{p,j})^2}{2\sigma^2} - \frac{1}{2} = \underbrace{\frac{1}{2} - \frac{1}{2}}_{=0} + \frac{(\mu_{q,j} - \mu_{p,j})^2}{2\sigma^2}.$$

Summing the squared differences over j produces $\|\boldsymbol{\mu}_q - \boldsymbol{\mu}_p\|_2^2$. □

Remark. This is the identity that makes diffusion models trainable. The objective is built out of KL divergences between distributions, which sounds hard; but the two distributions being compared have the *same* fixed variance, so the KL *is* a squared error on the means, up to the constant $1/2\sigma^2$. That is how a chain of KL terms becomes an ordinary regression loss, and eventually the familiar $\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\phi\|^2$.

Remark. The variances need not match exactly. If $q = \mathcal{N}(\boldsymbol{\mu}_q, \tilde{\beta}\mathbf{I})$ and $p = \mathcal{N}(\boldsymbol{\mu}_p, \sigma^2\mathbf{I})$, the same coordinatewise computation gives

$$\mathcal{D}_{\mathrm{KL}}(q \,\|\, p) = \frac{\|\boldsymbol{\mu}_q - \boldsymbol{\mu}_p\|_2^2}{2\sigma^2} + \frac{d}{2}\left[\frac{\tilde{\beta}}{\sigma^2} - 1 + \log\frac{\sigma^2}{\tilde{\beta}}\right].$$

The bracket contains no learnable parameter, so as a training objective it is an additive constant and may be discarded. This is what justifies using (6) even when the two variances are not equal.